What is a digital image...

- Digital images are visual representations of scenes or objects captured or generated in a form that a computer can store and process.
- Digital images are composed of discrete units called pixels as shown in the image below.
- Each pixel holds information about the image's color and brightness at a specific location, creating the overall visual structure when combined.



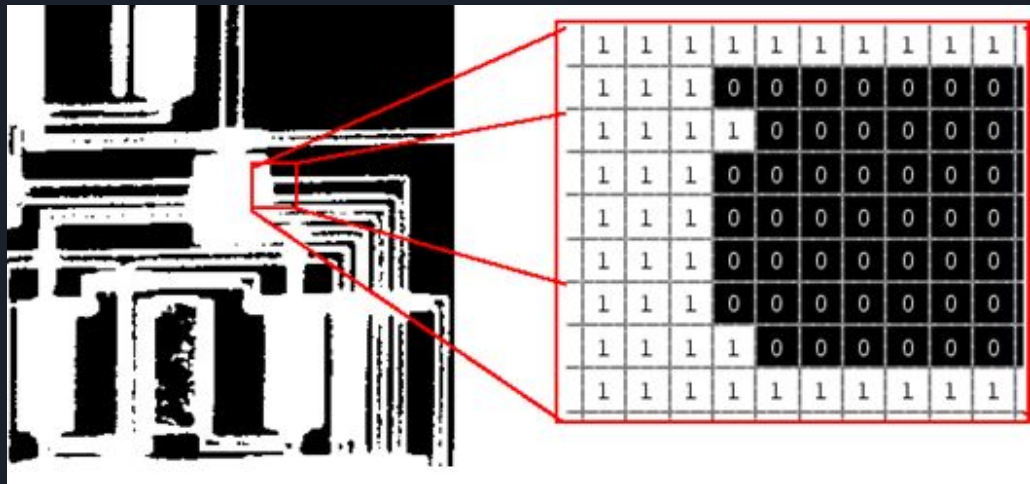
45	51	88	89	94	100	98	103	104	104
47	146	102	100	118	183	125	101	99	100
34	135	33	32	53	88	73	34	29	30
48	84	39	63	55	25	33	32	31	31
151	43	114	151	152	135	134	129	134	165
208	115	35	33	36	39	39	72	93	176
210	171	39	34	39	40	109	86	77	208
209	175	40	39	37	53	90	39	80	222
200	185	49	38	35	75	72	45	90	197
66	85	39	35	33	52	86	49	49	83

Digital Image ⟶ Grid of Pixels



Gray scale images

Binary images

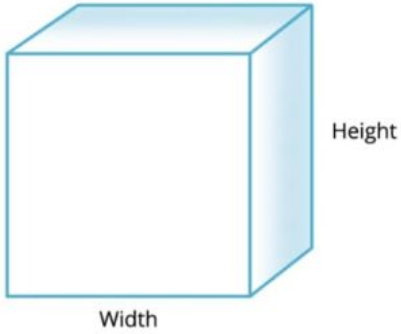


Color Image

Width x Height x Depth



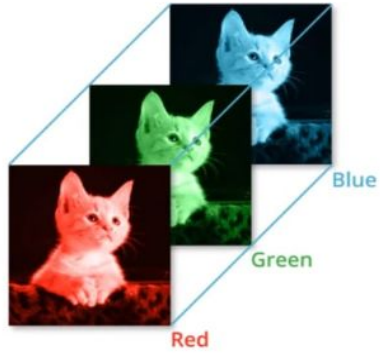
3D Array



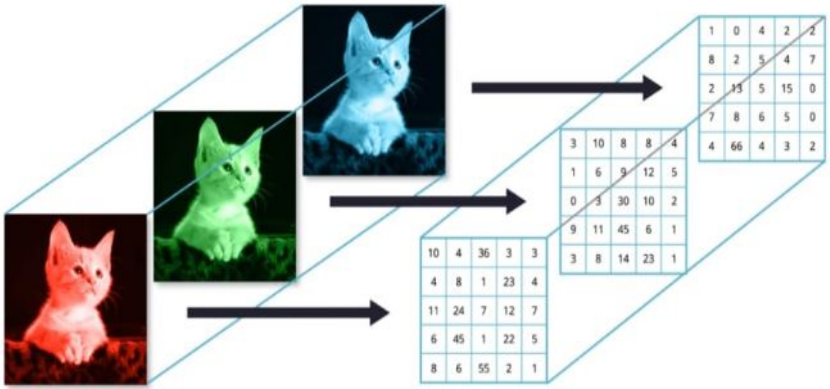
RGB Image

5 x 5 x 3





RGB Image

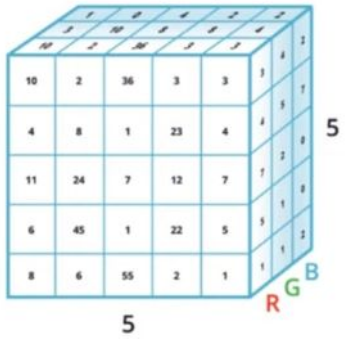


RGB Image

5 x 5 x 3



3D Array



8 Bit

Possible shade values per channel

256 x 256 x 256

R G B

16,777,216
Possible Colors



10 Bit

Possible shade values per channel

1,024 x 1,024 x 1,024

R G B

1,073,741,824
Possible Colors



12 Bit

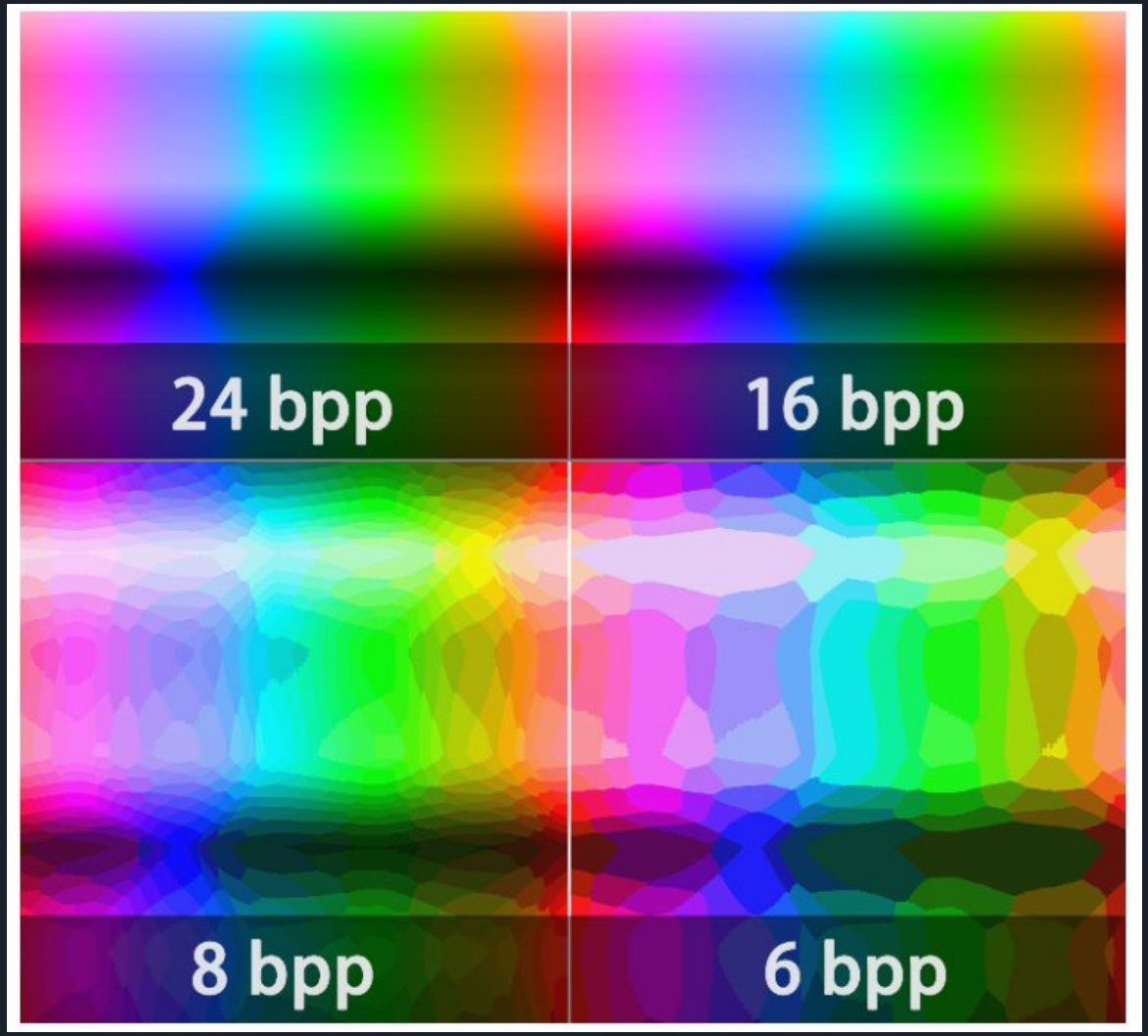
Possible shade values per channel

4,096 x 4,096 x 4,096

R G B

Over 68 Billion
Possible Colors

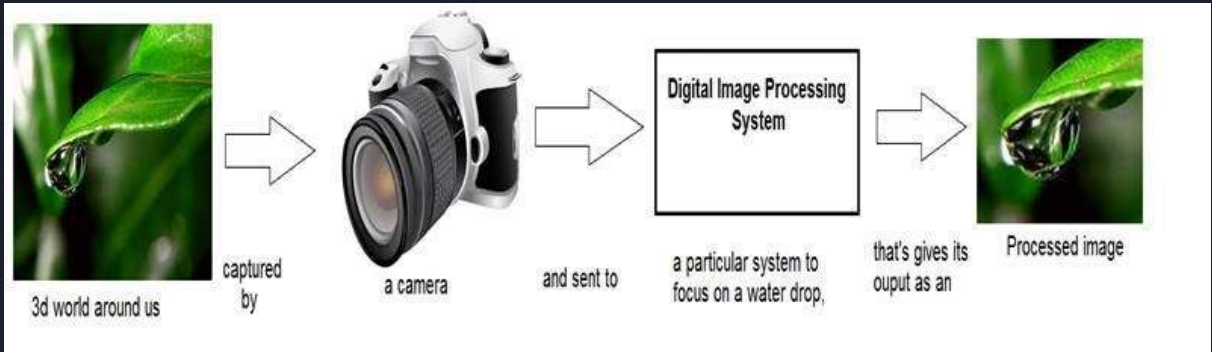



24 bpp
16 bpp
8 bpp
6 bpp

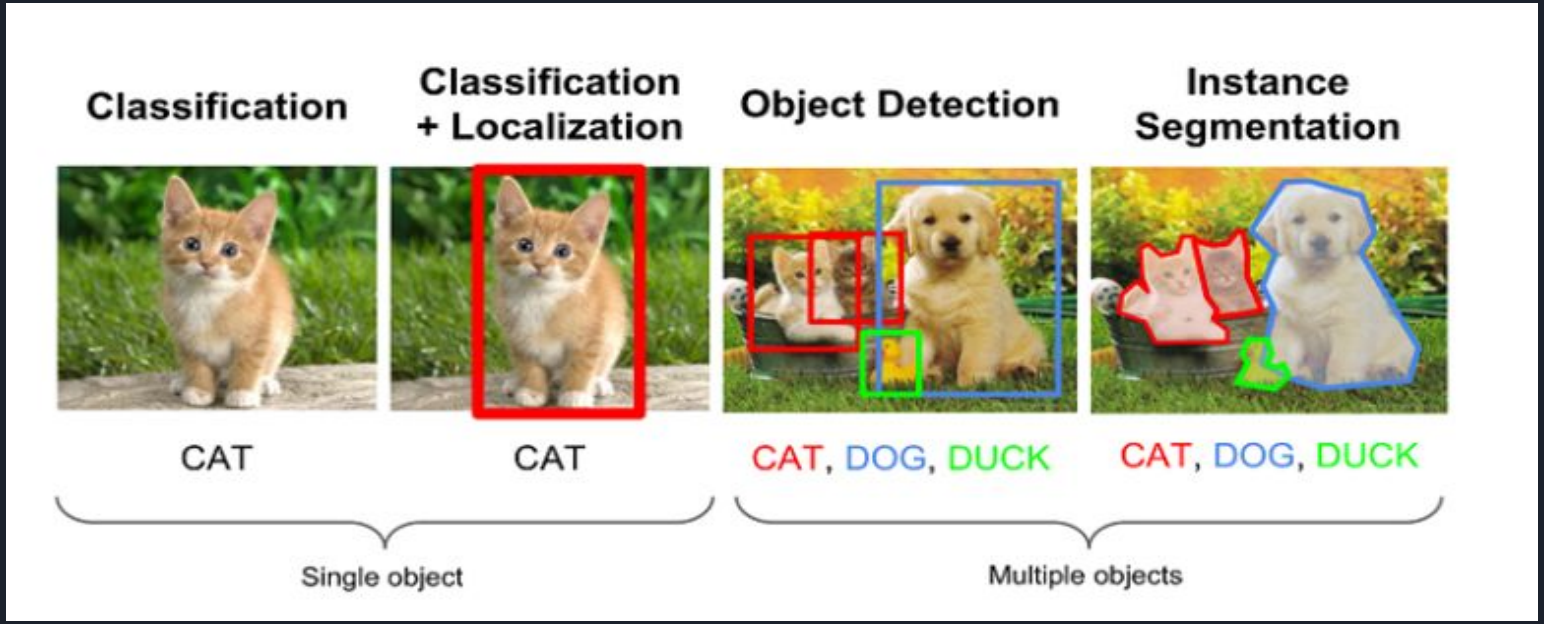
What is digital image processing and computer vision?

Algorithms, to process image

- Low level: sharpening, contrast, enhancement
- Mid level: image features are extracted
- High level: understanding, recognition






Classification
Classification + Localization
Object Detection
Instance Segmentation
CAT
CAT
CAT, DOG, DUCK
CAT, DOG, DUCK
Single object
Multiple objects


male, 55 years, laugh
female, 49 years, smile